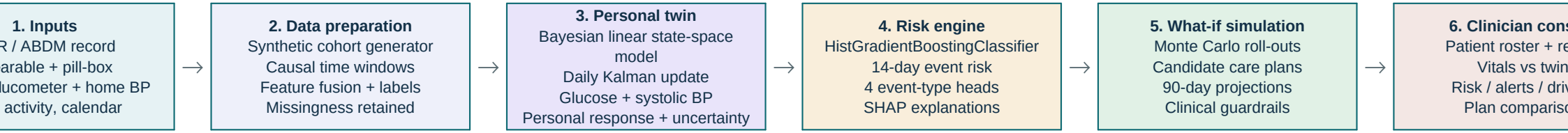
HridayTwin — System Architecture

Cardiometabolic digital-twin proof of concept | Winmasters | Goa Institute of Management



Data and evaluation loop: Synthetic patient streams are generated from hidden physiological states; the twin learns patient-specific response from noisy observations. Fused features feed the risk model, while the what-if engine rolls candidate plans forward and scores simulated futures using the risk engine.

Design principle: the twin simulates; the risk engine judges. Twin state features are intentionally excluded from the risk model because the documented ablation reduced AUROC.

Safety boundary: synthetic data only; research prototype, not a medical device, and not for clinical decision-making. Recommendations are for clinician review.